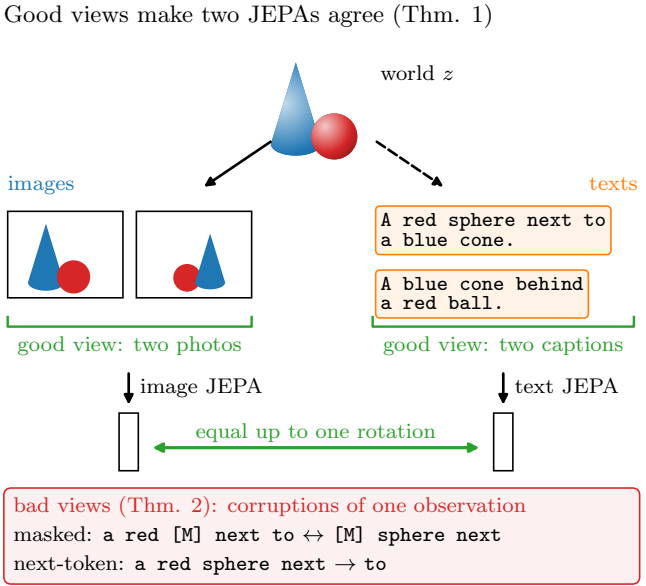
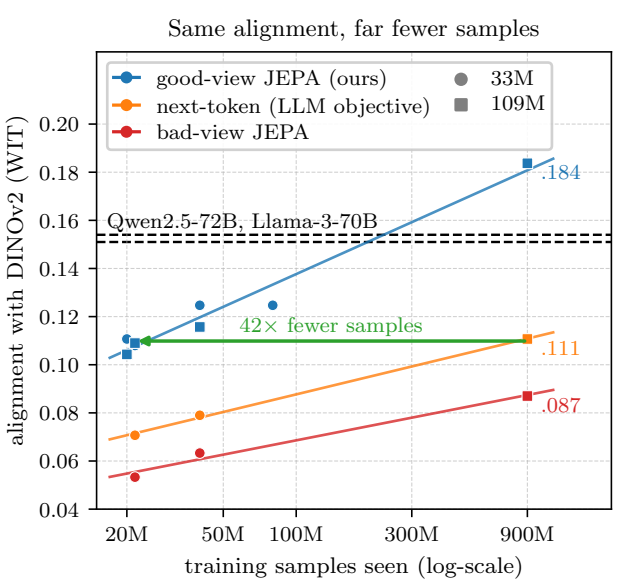

Good views make two JEPAs agree (Thm. 1)
world z
images
texts
A red sphere next to a blue cone.
A blue cone behind a red ball.
good view: two photos
good view: two captions
image JEPA
text JEPA
equal up to one rotation
bad views (Thm. 2): corruptions of one observation
masked: a red [M] next to ↔ [M] sphere next
next-token: a red sphere next → to


Same alignment, far fewer samples
good-view JEPA (ours)
next-token (LLM objective)
bad-view JEPA
33M
109M
alignment with DINOv2 (WIT)
0.20
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
Qwen2.5-72B, Llama-3-70B
42× fewer samples
.184
.111
.087
20M
50M
100M
300M
900M
training samples seen (log-scale)